BOROUGH OF MANHATTAN COMMUNITY COLLEGE
City University of New York

Department of Computer Information Systems
Office F930/Phone: 212-220-1476

Title of Course: **Introduction to Programming** **Class Hours: 3 Credits: 4**
CSC 111 – 0900 Fall 2025
Wed/Fri 9:00AM-11:15AM, Fiterman Hall, FH-907 **Laboratory Hours per Week: 2**
Instructor: Prof. Jose Ramon Santos josantos@bmcc.cuny.edu **Office Hours: TTH 4-5:30PM**

Course Description :

This course is an introduction to the fundamental concepts and terms of computer science, including algorithms, problem solving techniques, data types, concept of loops, conditional statements, modular programming, pointers, arrays, strings, basic file processing, structures and simple classes. Student will use a high-level computer programming language to solve a variety of problems.

Prerequisites:
Basic skills- ENG 088; ESL 094; ACR 094; MAT 206; CSC 101 or departmental approval

Course Student Learning Outcomes (Students will be able to…)	**Measurements (means of assessment for student learning outcomes listed in first column)**
1. Understand and apply the basic programming constructs.	**1.** Homework and programming projects.
2. Apply an appropriate problem solving techniques for developing an algorithmic solution to a problem.	**3.** Exam questions and programming projects.
3. Identify and provide solutions to common errors in programming by tracing the execution of a computer program.	**4.** Programming projects and exam questions.
4. Analyze and evaluate a computer program by applying the concept of debugging and testing.	**5.** Programming projects and lab exercises.
5. Develop a programming project by applying appropriate programming constructs.	**6.** Programming projects.

Below are the college's general education learning outcomes, the outcomes that are checked in the left-hand column indicate goals that will be covered and assessed in this course. (Check at least one.)

	General Education Learning Outcomes	**Measurements (means of assessment for general education goals listed in first column)**
☐	**Communication Skills-** Students will be able to write, read, listen and speak critically and effectively.	Essay questions and term project/paper.
☒	**Quantitative Reasoning-** Students will be able to use quantitative skills and the concepts and methods of mathematics to solve problems.	Use formulas and concepts of mathematics to solve problems in programming assignments.
☐	**Scientific Reasoning-** Students will be able to apply the concepts and methods of the natural sciences.	
	Social and Behavioral Sciences- Students will be able to apply the concepts and methods of the social sciences.	
	Arts & Humanities- Students will be able to develop knowledge and understanding of the arts and literature	
☒	**Information & Technology Literacy**- Students will be able to collect, evaluate and interpret information and effectively use information technologies.	Use a high-level computer programming language to create application software.
	Values – Students will be able to make informed choices based on an understanding of personal values, human diversity, multicultural awareness and social responsibility.	

Required Text & Readings

Textbook: **Starting out with C++ from Control Structures through Objects, 9/E**
Author: **Tony Gaddis**
Publisher: **Pearson**
ISBN: **978-0-13-4498379**

Other Resources : Flash drives are required.

Evaluation & Requirements of Students:

Exam 1	**20%**
Exam 2	**20%**
Final Exams	**30%**
Homework-Projects	**30%**
	100%

Outline of Topics:

Students should read the text section of the following chapters as well as the exercises. Reading these sections will help you prepare for your homework and examinations. Your instructor will be assigning specific exercises and programming problems from each chapter as homework.

Module Dates	Chapter	Topic	Assignments (Subject to change)
Module 1	Chapters 1 & 2	Introduction to Computers and Programming, IDE & Introduction to C++	IDE Setup and Pay program Pr1-1.cpp, Pr2-10, Pr2-21, Pr2-28
Module 2	Chapters 3 & 4	Expressions and Interactivity & Making decisions	Pr3-1, Pr3-3, Pr3-5, Pr3-13, Pr3-17, Pr3-19, Jupiter and Saturn, Pr4-2, Pr4-8&9, Pr4-14, Pr4-22, Crates
Module 3	Chapter 5	Loops and Files	Pr5-4, Pr5-7, Pr5-14, Pr5-24 Statistics
Module 4	Chapter 6	Functions (6.1-6.15)	Pr-6-5, Pr6-8, Pr6-11, Pr6-12, Pr6-16, Pr6-19, Pr6-22, Pr6-27 Employee Hours
Module 5		Review and Exam 1	
Module 6	Chapter 20	Introduction to Recursion (20.1 & 20.2)	Pr20-1, Pr20-2, Pr20-3
Module 7	Chapter 7	Arrays (7.1-7.4)	Pr7-17, Pr7-16, Pr7-19
Module 8	Chapter 7	Arrays (7.5-7.7)	Pr7-19, Rainfall
Module 9	Chapter 8	Searching and Sorting Arrays (8.1 & 8.3)	Pr8-1, Pr8-2, Pr8-3, Pr8-4, Pr8-5, Binary Search
Module 10	Chapter 9	Pointers (9.1-9.6)	Pr9-3, Pr9-6, Pr9-9, Pr9-10, Temp parameters, Case Study
Module 11		Review and Exam 2	
Module 12	Chapter 11	Structured Data (11.1-11.8)	Pr11-1, Pr11-2, Pr11-3, Pr11-4, Pr11-5, Pr11-6, Pr11-7, Account
Module 13	Chapter 13	Introduction to Classes (13.1-13.5)	Pr13-1, Pr13-2, Pr13-3, Pr13-4, Pr13-5, Rectangle Class
Module 14	Chapter 10	C-Strings and the String Class (10.1-10-5) More on Strings functions	Conversion of Strings Project
Module 15		Review and Final Exam	

Class Participation
Participation in the academic activity of each course is a significant component of the learning process and plays a major role in determining overall student academic achievement. Academic activities may include, but are not limited to, attending class, submitting assignments, engaging in in-class or online activities, taking exams, and/or participating in group work. Each instructor has the right to establish their own class participation policy, and it is each student's responsibility to be familiar with and follow the participation policies for each course.

BMCC is committed to the health and well-being of all students. It is common for everyone to seek assistance at some point in their life, and there are free and confidential services on campus that can help.

Single Stop www.bmcc.cuny.edu/singlestop, room S230, 212-220-8195. If you are having problems with food or housing insecurity, finances, health insurance or anything else that might get in the way of your studies at BMCC, come by the Single Stop Office for advice and assistance. Assistance is also available through the Office of Student Affairs, S350, 212-220- 8130.

Counseling Center www.bmcc.cuny.edu/counseling, room S343, 212-220-8140. Counselors assist students in addressing psychological and adjustment issues (i.e., depression, anxiety, and relationships) and can help with stress, time management and more. Counselors are available for walk-in visits.

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